

Program Question_5

```
Implicit none
Integer::n,r
Integer::result_npr,result_ncr,r1,r2

!calculate and Display the result
Write(*,*)"Enter the value of n"
Read(*,*)n
Write(*,*)"Enter the value of r"
Read(*,*)r

result_npr=nPr(n,r)
Write(*,*)"10P6 =",result_npr

Write(*,*)"Enter the value of r1"
Read(*,*)r1
result_ncr=ncr(n,r1)
Write(*,*)"10C4 =",result_ncr

Write(*,*)"Enter the value of r2"
Read(*,*)r2
result_ncr=ncr(n,r2)
Write(*,*)"10C6 =",result_ncr

Contains
!Using Recursive Function to calculate factorial
Recursive Function Factorial(n) Result(fact)
Integer,Intent(In)::n
Integer::fact
If(n==0 .or. n==1)Then
    fact=1
else
    fact=n*factorial(n-1)
End If
End Function
!Function to calculate npr=n!/(n-r)!
Function nPr(n,r) Result(cal)
Integer,Intent(In)::n,r
Integer::cal
cal=factorial(n)/factorial(n-r)
End Function

!Function to calculate npr=n!/(r!*(n-r)!)
Function nCr(n,r) Result(cal)
Integer,Intent(In)::n,r
Integer::cal
cal=factorial(n)/(factorial(r)*factorial(n-r))
End Function

End Program
```